INT'L. J. PSYCHIATRY IN MEDICINE, Vol. 18(3), 1988

THE DIAGNOSTIC DILEMMA OF MYXEDEMA AND MADNESS, AXIS I AND AXIS II: A LONGITUDINAL CASE REPORT*

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ABSTRACT

A patient with presumed chronic paranoid schizophrenia had chronic thyroiditis and Grade I hypothyroidism. Psychosis cleared following treatment with thyroid replacement. The probable presence of two axis II disorders may have contributed to the missed medical diagnosis and the patient's eventual suicide. The personality disorders were a major problem in the patient's medical and psychiatric care. The differential diagnosis among hypothyroidism and primary axis I psychotic and depressive psychopathology has always been problematic. When axis II pathology is also present, the diagnostic dilemma is increased.

Myxedema is commonly used as a synonym for advanced hypothyroidism [1]. The popular term "myxedema madness" was coined in 1949 [2] and refers to the common association of myxedema and serious psychiatric symptoms [3,4]. Both depressive affect and paranoid ideas, delusions and suspiciousness are common. Other features commonly found in hypothyroid psychosis include

*Supported in part by the San Diego Veterans Administration Medical Center Research Service and by the NIMH Mental Health Clinical Research Center (MH 30914).

doi: 10.2190/Y6YM-9F5W-D24L-34AK
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suicidal ideation and attempts, auditory or visual hallucinations (often both together and quite vivid), olfactory or tactile hallucinations, delusions of many types, irritability and intermittent rages, mood swings, and hyperactivity. We present a case of myxedema madness admitted to the Mental Health Clinical Research Center (MHCRC) at UCSD School of Medicine. Prior to admission, the patient had the diagnosis of chronic paranoid schizophrenia, a diagnosis which we could not make given the exclusion criterion "F" [5] for schizophrenia, "Not due to any Organic Mental Disorder . . .".

CASE REPORT

Mr. A, a twenty-four-year-old, never married, unemployed Caucasian male, was admitted to the Special Evaluation and Treatment Unit and NIMH Mental Health Clinical Research Center (MHCRC) of the San Diego Veterans Administration Medical Center (SDVAMC) and UCSD School of Medicine for suicidal ideation and paranoid, delusional thinking. At age nineteen, while he was in the military, he was hospitalized after striking shore patrol officers with a knife when seeing them "turn into rats," A diagnosis of schizophrenia was made, he was treated with haloperidol with minimal success and transferred to a hospital near his home. In this second hospital stay a diagnosis of mixed substance abuse was made, and the patient was discharged from the military. During the three years prior to the index admission, the patient occasionally heard voices, had periods of delusional thought and attempted suicide by electrocution, overdose and wrist slashing. Thyroid function was not evaluated during any of three previous hospitalizations.

Developmental history revealed that the patient met DSM-III criteria for a conduct disorder of childhood, and had more than three of the needed criteria before age fifteen to meet criterion "B" [5] for Antisocial Personality Disorder. Medical history revealed cold intolerance; chronic fatigue; lethargy; constipation; a thirty-pound weight gain in five years; low, hoarse voice; dry hair; flaking skin; delayed puberty; and recent impotence. Past medical history revealed no previous history or family history of thyroid disease. Historically he had been a binge drinker, though he had not binged on alcohol recently. He also had a history of extensive use of marijuana, PCP, and LSD while in high school. Family history included many relatives in his mother's family who attempted suicide, and possible psychiatric illness in a sister.

Admitting physical examination revealed a moderately overweight Caucasian male who did not appear acutely ill with slow, monotonic speech; bradycardia (pulse rate = 40); dry, pale, cool skin; and dry, sparse hair. Admitting mental status examination revealed logical thought processes, blunted though saddened affect, paranoia, delusions, grandiosity, auditory hallucinations, ideas of reference, and suicidal ideation. On Mini-Mental Status Examination,

Table 1. Laboratory Results

	Admitting	*After 7 weeks L-thyroxine Replacement*	*Reference Ranges*
T4	<2.5 mcg/dl		4.5-11.5 mcg/dl
TSH	353 uIU/1	16 uIU/1	0-7 uIU/1
T3RU	24%		30-40%
hemoglobin	12.7 gm/dl	14.3 gm/dl	14-18 gm/dl
Fe	13 ug/dl		65-175 ug/dl
Ferritin	187		
fasting blood sugar	69 mg/dl	106 mg/dl	70-110 mg/dl
mean red cell volume	96.1 cmu	91.1 cmu	82-92 cmu
SGOT	101 IU/L	21 IU/L	10-45 IU/L
SGPT	82 IU/L	42 IU/L	10-45 IU/L
CPK	2546 IU/L		0-175 IU/L
LDH	418 IU/L	152 IU/L	25-200 IU/L
serum cholesterol	379 mg/dl	260 mg/dl	150-250 mg/dl
testosterone	561 ng/dl		250-1000 ng/dl
LH	7.8 mIU/ml		0-20 mIU/ml
MICR AB	1:25,600 dilution		< 1:100 dilution
THYR AB	>1:5120 dilution		< 1:125 dilution
VDRL	Not Reactive		

he scored 29 out of a possible 30, demonstrating minimal cognitive deficit to initial clinical testing. He also met DSM-III criteria for antisocial personality disorder on axis II, with chronic inability to sustain consistent work behavior, failure to accept social norms with respect to lawful behavior, irritability and aggressiveness, and impulsiveness. However, in the context of chronic hypothyroidism of unknown duration, the etiology of these traits was unclear.

The laboratory findings (See Table 1) showed that both thyroxine (T4) and triiodothyronine by radioactive uptake (T3RU) were low, while thyroid stimulating hormone (TSH) was elevated. Hemoglobin, serum iron, and fasting blood sugar were low, and red cell volume, serum glutamic oxalacetic transaminase (SGOT), serum glutamic pyruvic transaminase (SGPT), creatinine phosphokinase (CPK), lactate dehydrogenase (LDH), and cholesterol were elevated. Further examination of the patient revealed a slight asymmtery of the thyroid (R > L) and hyporeflexia of the ankle jerks bilaterally. Admitting toxicology screen was negative for drugs of abuse, including alcohol. Electrocardiogram showed sinus bradycardia, low amplitude, and right ventricular hypertrophy. Cephalic computerized axial tomography (CAT) scan and electroencephalogram were normal. Neuropsychological testing showed general intellectual functioning in the average range, intact fundamental verbal and visuo-spacial skills, above average memory for visuo-spacial material, with a mild problem in free recall of verbal material, normal recognition memory for verbal material, and normal concept formation skills.

Given a TSH > 100 (See Table 1), the endocrinology consultant diagnosed primary hypothyroidism, probably due to autoimmune chronic lymphocytic thyroiditis. Anti-microsomal antibodies were 1:25,600, confirming the etiology of the thyroid disease. The patient had been delusional with periods of loose and inappropriate thought patterns. Initial Brief Psychiatric Rating Scale (BPRS) [6] score was 56 (psychosis subscale = 26), Hamilton Rating Scale for Depression (HRS-D) [7,8] score was 21 on the 21 item test (HRS-D > 15 is consistent with major depression), and Biegel-Murphy [9] score was 29. The patient received Research Diagnostic Criteria (RDC) [10] and DSM-III [5] diagnoses of Other Psychiatric Disorder (Organic Delusional Disorder) and Antisocial Personality Disorder, based on a Schedule for Affective Disorders and Schizophrenia (SADS) structured interview and a diagnostic consensus meeting of three psychiatrists at our MHRC. L-thyroxine was started at a dose of 50 micrograms (mcg) orally daily, and slowly increased. Four days after initiation of replacement, he became more confused, labile and withdrawn with increased paranoid and delusional content to his thought. The subsequent increases in L-thyroxine replacement to the 100 and 150 mcg levels required management with haloperidol, benztropine and locked ward restriction because of increased paranoia and disordered and delusional thought. The delusions and paranoia slowly cleared. After several weeks of neuroleptic and L-thyroxine treatment the patient was discharged to a board and care facility without evidence of a formal thought disorder, with a BPRS score of 44 (psychosis subscale = 19), a HRS-D score of 19, and a Biegel-Murphy score of 11. Repeat laboratory examination at seven weeks revealed improvement of many of the abberant values (see Table 1). The patient's haloperidol was discontinued at this time.

At three months follow-up without haloperidol, the patient remained clear of a thought disorder on only L-thyroxine 200 mcg daily. It became obvious that his major depression had not cleared with successful treatment of his hypothyroidism, as we had expected, and desipramine was added to the L-thyroxine therapy. Clinically, his cognition was normal. During his post-discharge psychotherapy, the patient proved difficult. As expected with his antisocial personality disorder, he was impulsive and reckless. He developed an intense transference with his (female) psychiatrist, was quick to anger, manifested an identity disturbance (Who am I? How did I get here in my life? Where am I going in life? Who will help me find my way?), showed marked affective instability and chronic feelings of boredom. These problems gained clarity as his major depression cleared, and further history revealed that such traits had been typical of his long term functioning. A tentative additional axis II diagnosis of borderline personality disorder was made, but given the chronicity of his hypothyroidism, the etiology of these traits remained unclear.

During this period, his physiognomy and body habitus improved noticeably. He became thinner and more agile in his movements, his face was more expressive and less "puffy," his skin was more supple and his hair less dry and unkempt. His speech increased in speed to a more normal rate, and his voice was no longer monotonous. Given this obvious clinical improvement in his myxedema, the further expense of thyroid function studies was deemed unnecessary by our endocrinology consultant.

At four months follow-up, the patient continued to do well with respect to psychotic and depressive symptoms on L-thyroxine 200 mcg and desipramine 150 mg daily. No deficits in cognition were noted clinically. At this point the patient began missing appointments and stopped both medications. One month later he was readmitted to the inpatient unit, with a paranoid delusional system and vivid auditory and visual hallucinations. During this hospital stay, the haloperidol, L-thyroxin and desipramine were again started, and the diagnoses of major depression and tentative antisocial and borderline personality disorders were again made. As with his previous admission, the patient first worsened, then cleared. He was discharged with no evidence of a formal though disorder. At three months follow-up after this second admission, the patient was clear of psychosis without haloperidol and living in a board and care facility, was involved in outpatient psychotherapy for mild depression, and continued on only L-thyroxine 200 mcg and desipramine 150 mg daily. A higher desipramine dose was not used because the patient's mood was continuing to improve and he complained of medication side effects. There were no cognitive deficits noted clinically.

At six months follow-up after the second admission, while living at the board and care facility, the patient completed suicide with an overdose of medication. At the time of his suicide, the patient was receiving supportive psychotherapeutic care at three agencies. He had received $8,000.00 in back social support from a public agency, had paid all outstanding debts, had purchased a $3,000.00 motor bike, and was actively searching for an apartment for independent living. On mental status examination, thought content and process, mood, cognition, and perception were all reported to be normal, with no depression, delusional thinking or hallucinations. A fellow patient told clinicans at the board and care that the suicide had been planned secretly for about six months, that they were supposed to suicide together, but that he decided not to suicide.

DISCUSSION

Mr. A presented as a 24-year-old gentleman with complaints of depressive and psychotic symptomatology with a psychiatric history of five years duration, chronic grade 1 hypothryoidism of unknown duration and a history consistent with both antisocial and borderline personality disorder. His hospital and post hospital course revealed elimination of psychotic symptomatology with institution of L-thyroxine replacement, with return of psychosis when he stopped L-thyroxine for one month. Sustained improvement in thought content, process, and cognition seemed unrelated to anti-psychotic medication. The initial failure to make the medical diagnosis and his eventual successful suicide may both be attributable in part to the combination of traits consistent with severe axis II psychopathology.

The medical history and clinical findings are classic for Grade I hypothyroidism, supported by laboratory findings (See Table 1). Both T4 and T3RU were low, while TSH was dramatically elevated. Consistent with a diagnosis of hypothyroidism, hemoglobin, serum iron, and fasting blood sugar were low, and red cell volume, SGOT, SGPT, CPK, LDH, and cholesterol were elevated. (Increased concentrations of SGOT, SGPT, CPK, and LDH are a frequent but not invariable manifestation of the hypothyroid state.)

At three months follow-up desipramine was started because Mr. A's major depression had not cleared with successful treatment of his hypothyroidism. Desipramine was not started prior to this time, because the florid mental disturbances of hypothyroidism in the adult usually respond to adequate replacement with thyroid hormone [11, 12]. An exacerbation of the psychosis early in the pharmacologic treatment of hypothyroidism has long been reported [13]. This initial worsening with the beginning of thyroid supplementation is not uncommon [14]. The probable mechanism is that the rise in metabolic rate occurs before central nervous system function improves, so that initially the increased metabolic rate seems to drive the psychosis [15]. Because of this worsening, severely disturbed patients should be admitted to a hospital to begin thyroid supplementation [12].

When the tentative diagnosis of borderline personality disorder was made, an alternative diagnosis could have been a thyroxine-induced hypomania, which has been reported at least eighteen times in the literature [16]. However, at this time the patient clinically had symptoms more consistent with depression than hypomania, while his physiognomy and body habitus showed definite clearing of the myxedematous state.

The antisocial personality disorder has traits which began prior to age fifteen, as required by the DSM-III criteria for this illness. These traits were intermixed with symptoms of a conduct disorder of childhood. Since the date of onset of his hypothyroidism is unknown, the true etiology of these personality traits remains uncertain. The patient had been myxedematous for several years, possibly since his initial presentation in 1981. The possibility exists that the organic illness not only mimicked Axis I pathology, but also contributed to erroneous Axis II diagnoses of borderline and antisocial personality disorder.

As in previously reported studies, Mr. A's endocrinopathy was not evident on initial assessment in 1981 [17, 18]. A recent study showed that 12.2 percent of 270 patients presenting with a chief complaint of depression had clinical and laboratory findings suggesting hypothyroidism [19]. We were unable to locate a similar study of patients presenting with psychosis. This case illustrates the need to evaluate organic, treatable bases for presenting psychopathology despite a long prior psychiatric history or underlying personality disorders which may obscure the fact that organic illness is mimicking Axis I or Axis II pathology. This report also raises the question of frequency of clinical hypothyroidism in the population diagnosed with any of the chronic schizophrenias, a question unanswered in the existing literature.

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